



RUTGERS
BIOMEDICAL AND
HEALTH SCIENCES

AI-driven discovery in biomedicine

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Data Science and AI Revolution

- Google: “taking a fundamentally new and

Former Postdocs/Graduate Students from the Johnson Lab:

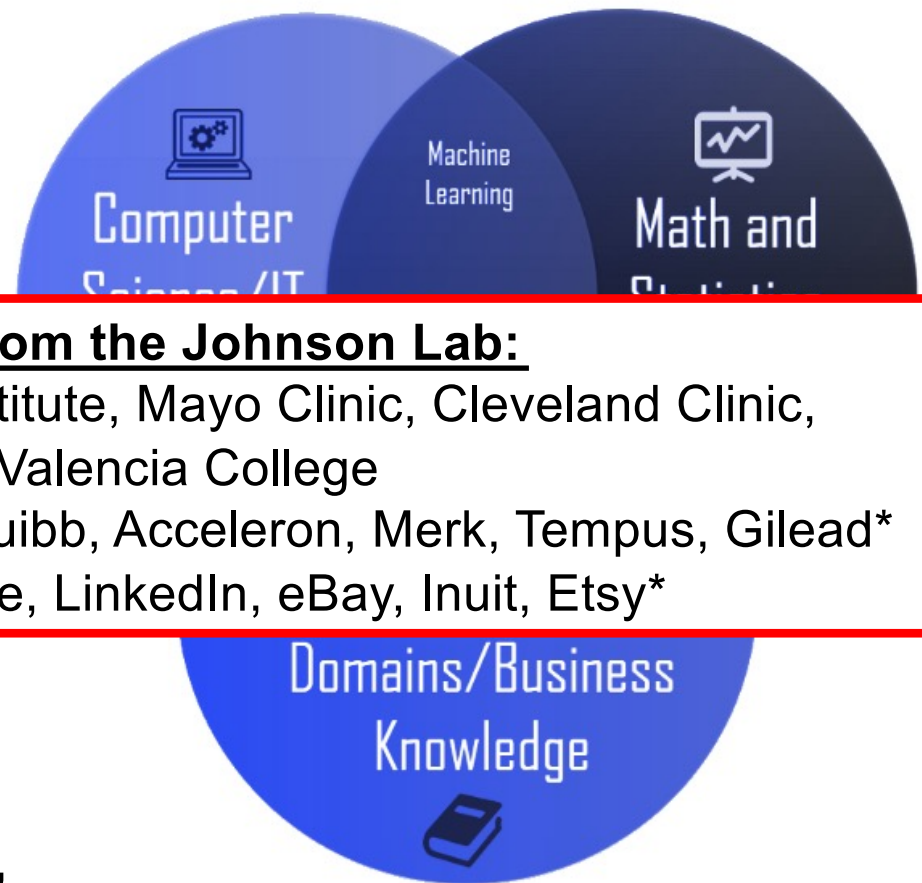
Academics/bio research: MIT/Broad Institute, Mayo Clinic, Cleveland Clinic, BYU, Xi'an Jiaotong-Liverpool University, Valencia College

Pharma: AstraZeneca, Bristol Meyers Squibb, Acceleron, Merck, Tempus, Gilead*

Data Science: Google, Amazon, YouTube, LinkedIn, eBay, Inuit, Etsy*

and data better! Data is the lifeblood of science.

- Pedro Domingos: "**AI will not replace us, it will amplify us.**"



Machine Learning and Artificial Intelligence

- **Machine learning (ML)** is a computer's way of learning from examples, and it is one of the most useful tools we have for the construction of *artificially intelligent* systems
- **Artificial intelligence (AI)** is a term used when machines or computers can mimic functions that humans can do.



To Categories of Machine Learning (ML)

Supervised: Outcomes (labels) are known

- Goal: *Predict an outcome*
- Examples:
 - Classification (e.g., disease vs. no disease)
 - Regression (e.g., predicting cost, risk, or score)

Unsupervised: Outcomes are not known

- Goal: Discover structure in data
- Examples:
 - Clustering (finding groups or subtypes)
 - Dimensionality reduction (simplifying complex data)

Common ML Algorithms

Supervised Algorithms

- Linear / Logistic Regression
- Decision Trees & Random Forests
- Support Vector Machines (SVM)
- Neural Networks / Deep Learning

Unsupervised Algorithms

- K-means & Hierarchical Clustering
- DBSCAN
- Principal Component Analysis (PCA)
- Autoencoders

*With great power
comes great
responsibility.*

-Uncle Ben,
Spiderman

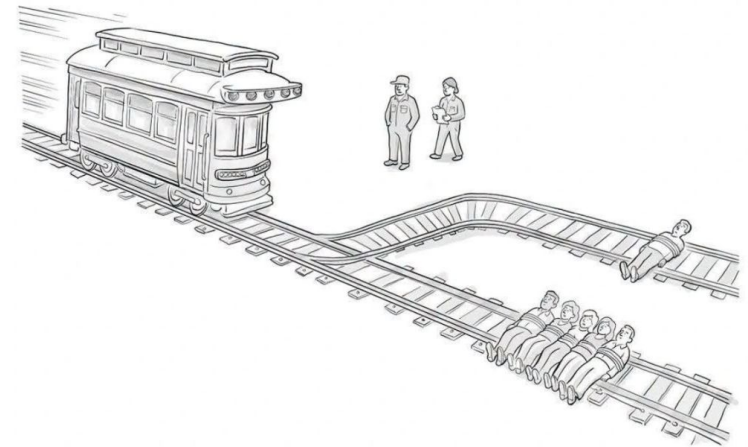


The New Yorker · Follow

1d ·



Today's Daily Cartoon, by Paul Noth.



PAUL
NORTH

*"Luckily, the A.I. trolley will
decide who to kill all on its own."*



Here is a picture of a basketball player and another of me and my wife. Can you make me a basketball player on the Boston Celtics?

Using AI: Images

Here is a picture of a basketball player and another of me and my wife. Can you make me a basketball player on the Boston Celtics?



This is great. Make my wife on the jazz and have me dunking on her



You already use AI every day

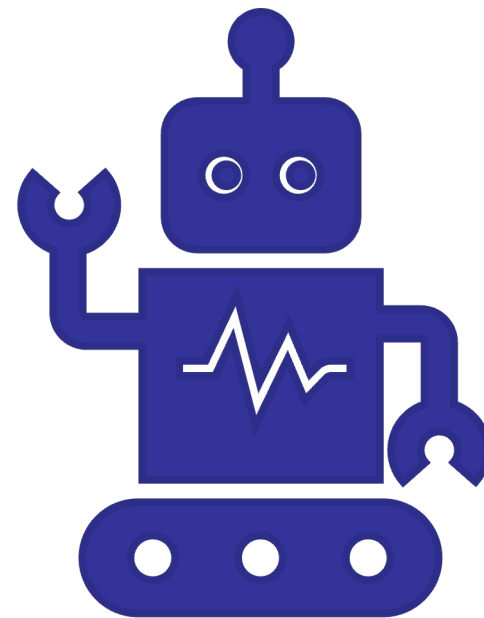
- Who uses Snapchat filters?
- Spotify or YouTube recommendations?
- Google Maps?
- Siri, Alexa, or ChatGPT?
- Video games where the computer adapts to you?



Narrow AI (All AI today)

AI designed to do one specific task well.

- Face ID
- Recommendation systems
- Google Maps
- Spam filters
- AI in games
- ChatGPT



Predictive AI (application of ML)



AI designed to do forecast and classify



Analyzing historical data to predict or categorize future outcomes



Regression, random forests, support vector machines, neural networks

Generative AI (GenAI)

AI that creates new content based on patterns it has learned.

- Writing text (ChatGPT)
- Creating images
- Music generation
- Video and voice generation

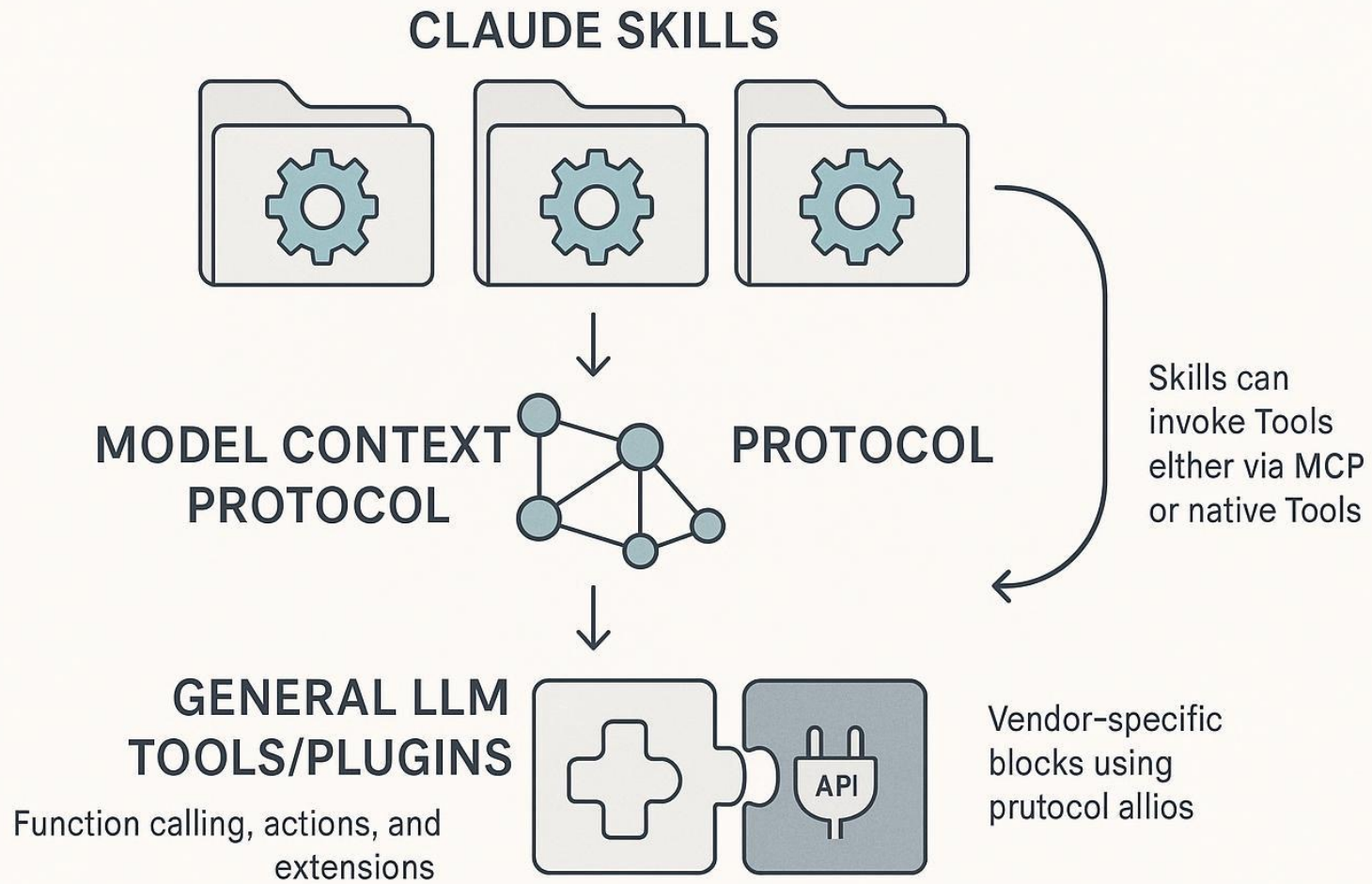


Agentic AI (emerging, very relevant)

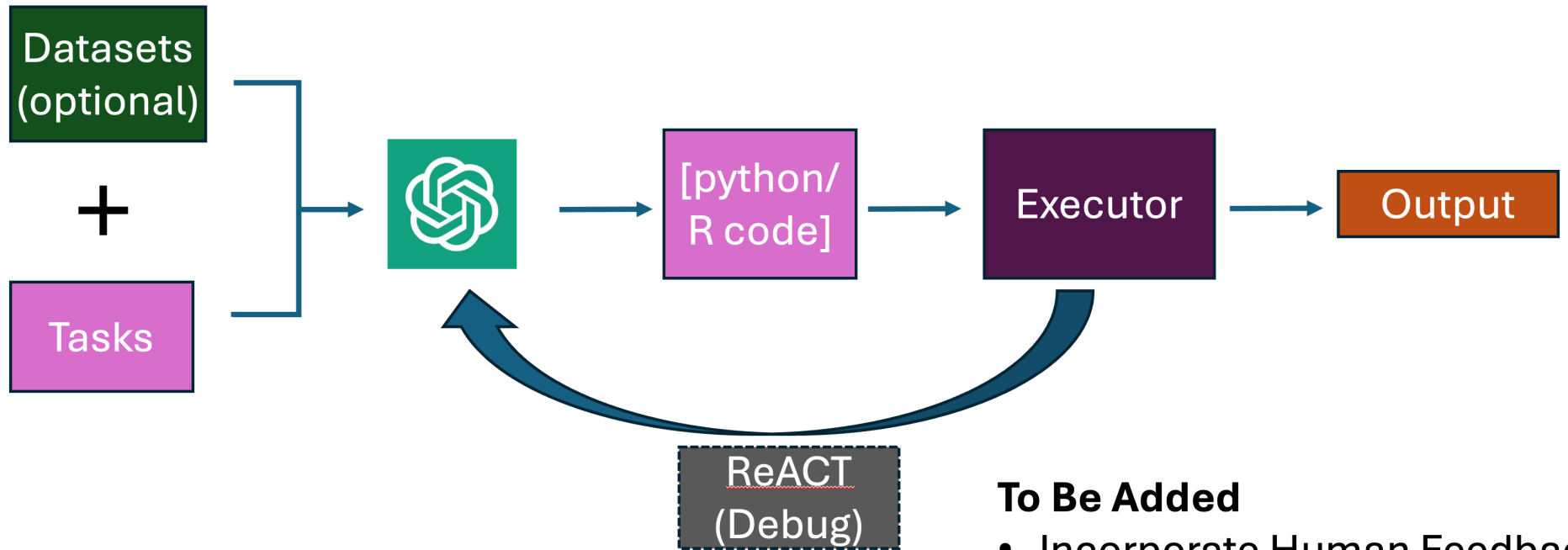
AI systems that can take actions, make plans, and use tools to accomplish goals.

- AI that:
 - Schedules meetings
 - Writes emails and sends them
 - Searches the web and summarizes
 - Runs tasks automatically





Inside the Agent



Current Bottleneck

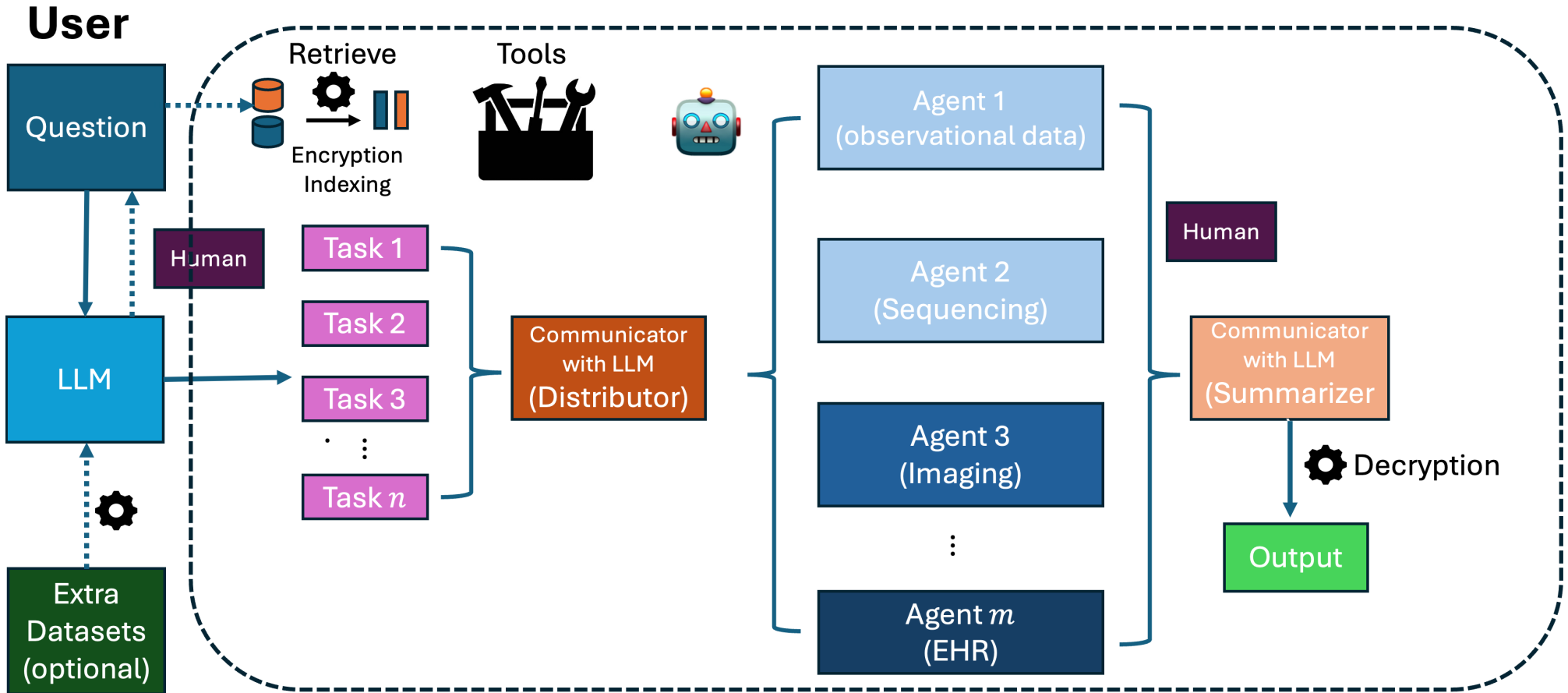
- Requires scalable multi-node parallelism

To Be Added

- Incorporate Human Feedback
- Memory Integration
- RAG
- More to come

Use Enhanced LLM to Build AI Agent

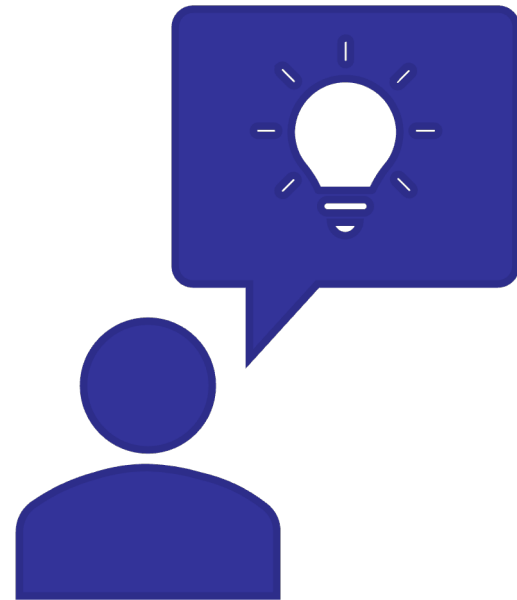
Docker



General AI (AGI) — does NOT exist

AI that could think and reason like a human across many tasks.

- This is theoretical
- It does not exist
- Movies exaggerate it



Comparison

Feature	Predictive AI	Generative AI	Agentic AI
Primary Goal	Forecast accuracy & Classification	Content creation & Human-like response	Task completion & Autonomous action
Interaction	Input → Class/Value	Prompt → Response (Chat)	Goal → Multi-step Workflow
Output	Numbers, Labels, Probabilities	Text, Images, Audio, Code	Completed Tasks, API Calls, File Edits
Reasoning	Statistical correlations	Pattern matching & Probability	Planning, Reflection, & Tool Selection

Real Intelligence (Human Intelligence)

The ability to think, choose, create, understand meaning, and act with moral responsibility.

- Agency – we choose
- Moral judgment – right vs. wrong
- Creativity – truly new ideas
- Understanding meaning
- Empathy and relationships
- Spiritual capacity

*AI will not replace us,
it will amplify us.*

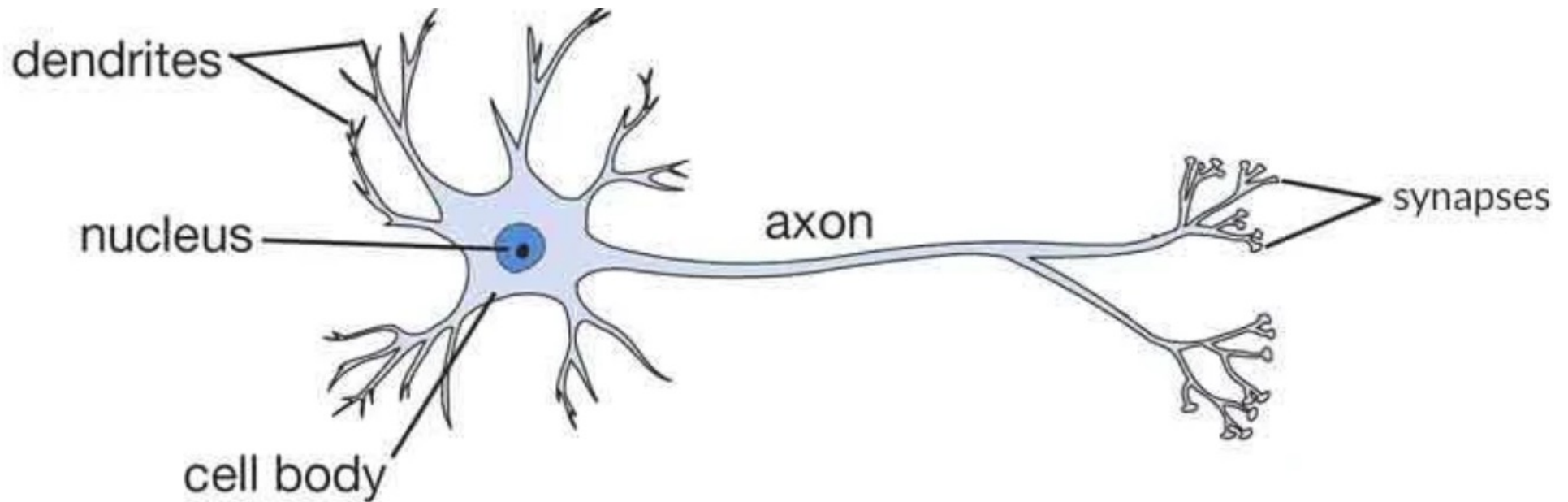
-Pedro Domingos

<https://www.instagram.com/reels/DSsE2dXE1zt/>

AI is not magic; it's math.

-Peter Norvig

Biological Nerve Cells

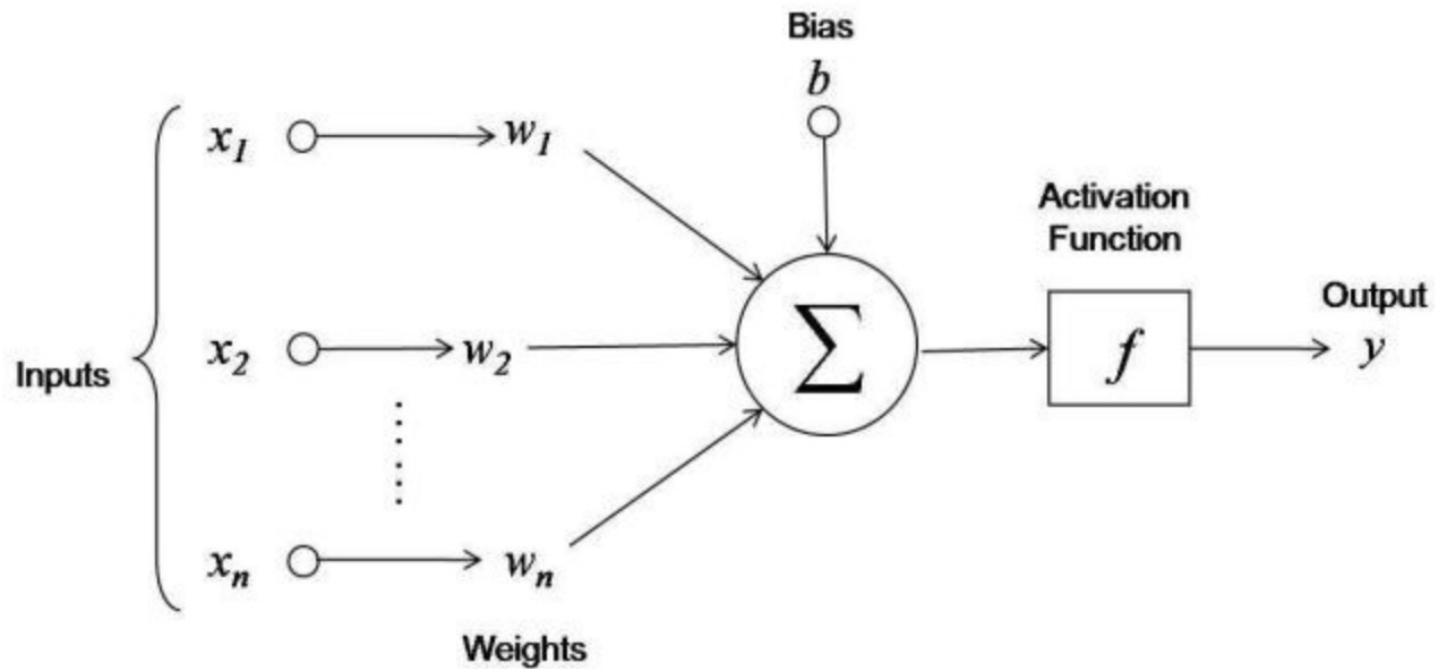


Neural Networks

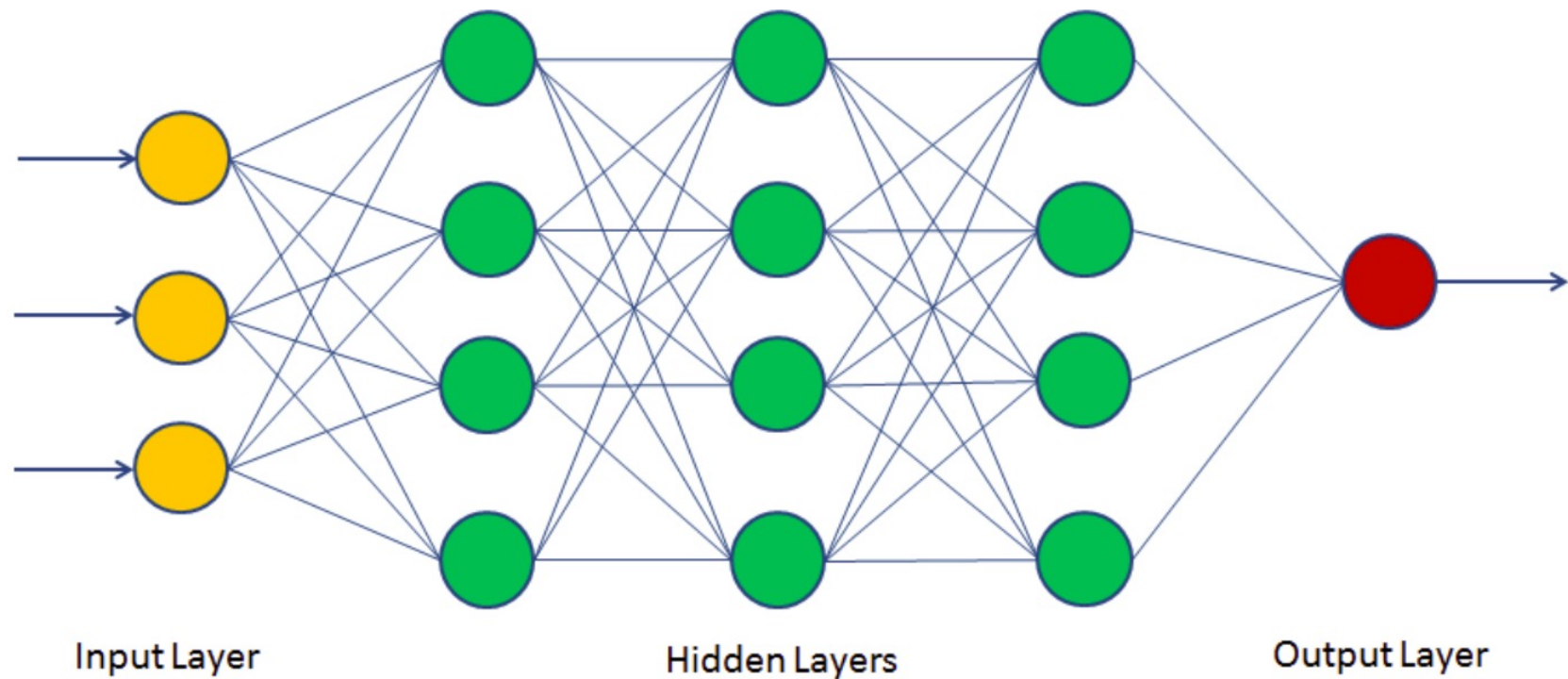
- The human brain consists of billions of neural cells that process information.
- Each neural cell is a simple processing system.
- The interconnected web of neurons, a **biological neural network**, transmits information through electrical signals from neuron to neuron.



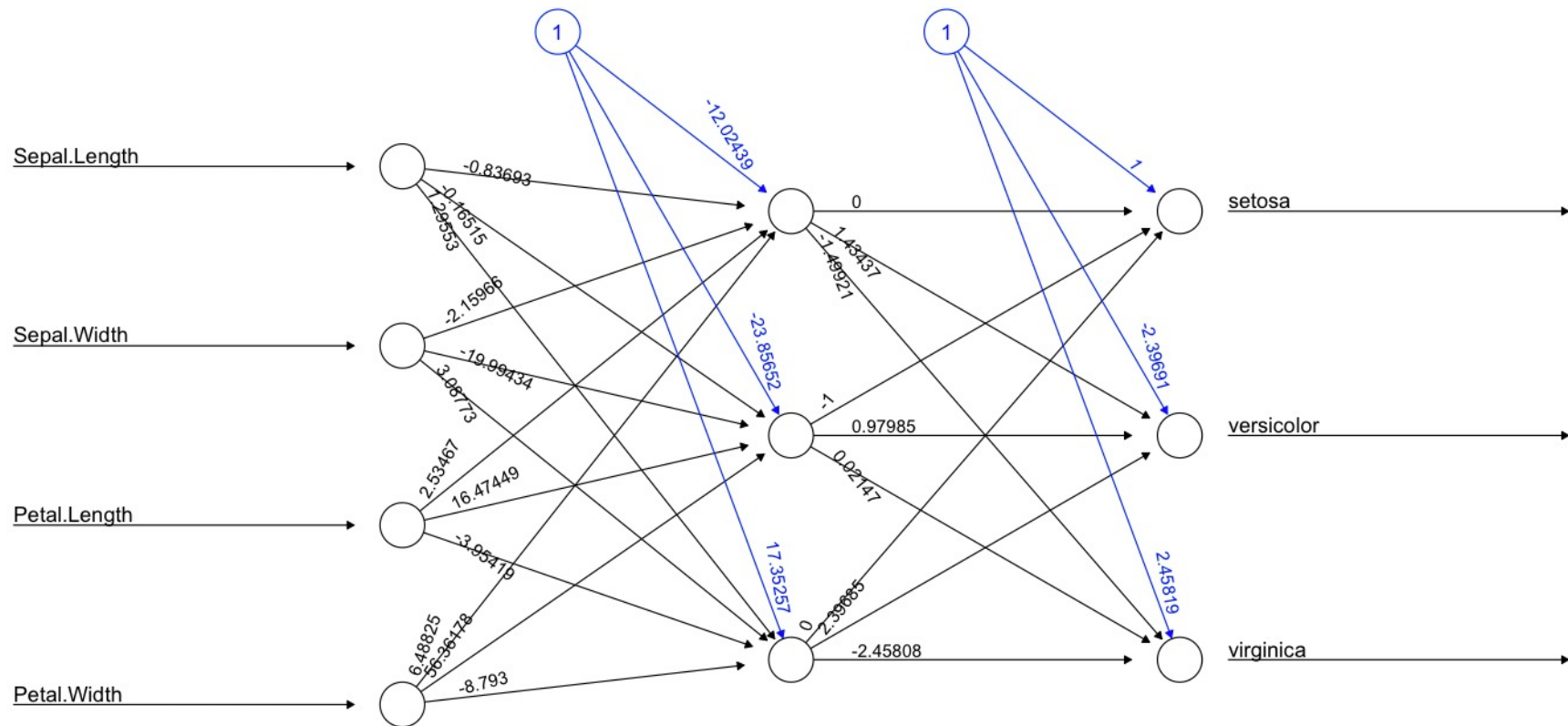
The Perceptron model (1957) looked like this:



Modern Neural Networks: Multi-layer Perceptrons



Modern Neural Networks: Multi-layer Perceptrons



Error: 1.934716 Steps: 26654

Chain Rule: When will I ever use this??

**The Famous
CHAIN RULE**

$$\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x)$$

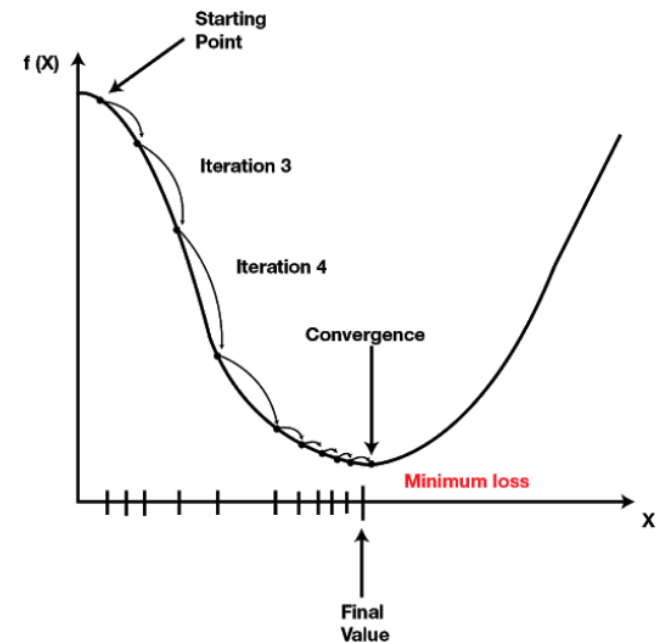
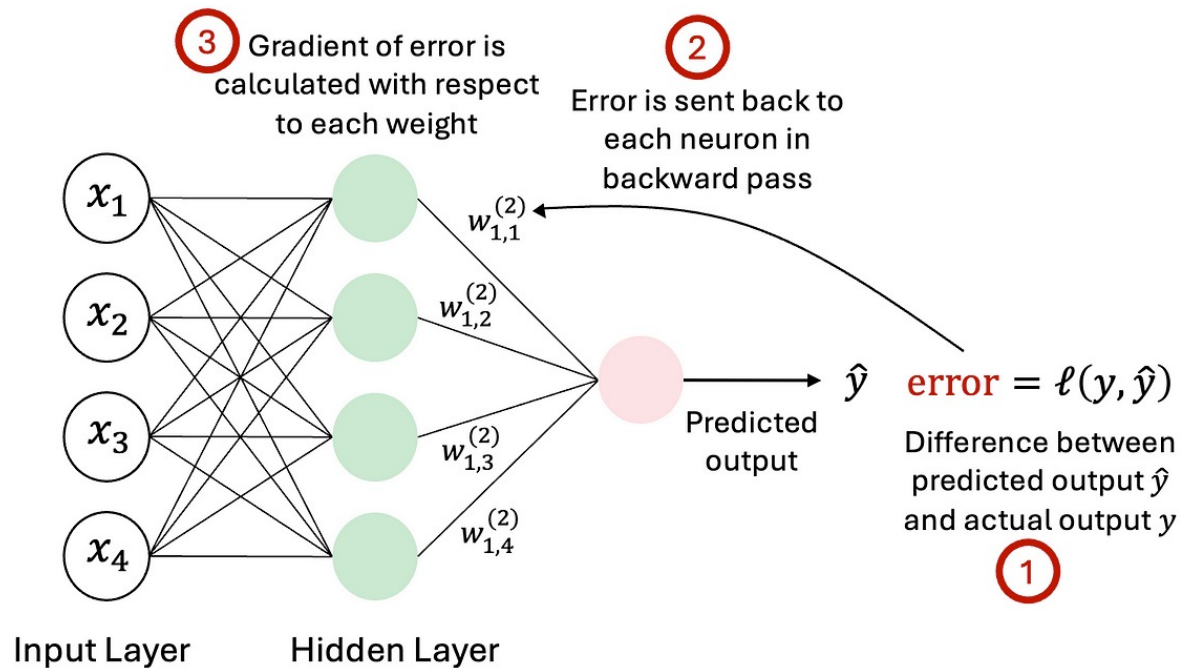


Chain Rule: When will I ever use this??

$$f' \circ g \circ h \circ k(x) \quad \times \quad g' \circ h \circ k(x) \quad \times \quad h' \circ k(x) \quad \times \quad k'(x)$$

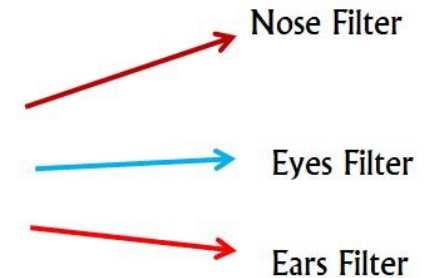
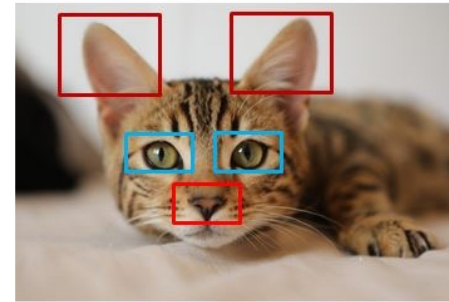


Chain Rule: Backward Propagation in Neural Nets



Specialized neural network (deep learning)

- Pattern recognition
- Grid-like data (images)
- Learning features
- Apply filters (kernels)



Cat



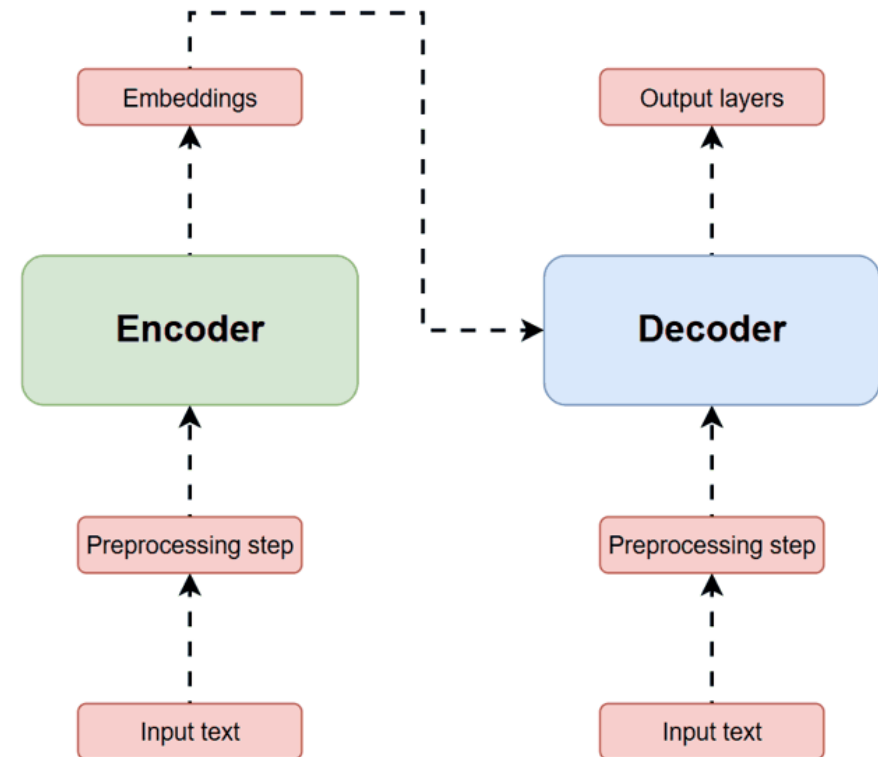
Not a cat

Transformer models: Language prediction

“GPT” = Generative Pretrained Transformer Model

1. You type a prompt
2. The model:
 - Looks at all the words
 - Assigns probabilities to possible next words
3. It picks the most likely next word
4. Then repeats—again and again

ChatGPT is a very advanced autocomplete.



AI should never replace human intelligence—it should enhance it

Examples:

- Doctors + AI = better diagnoses
- Teachers + AI = more personalized learning
- Engineers + AI = safer systems
- Leaders + AI = better information



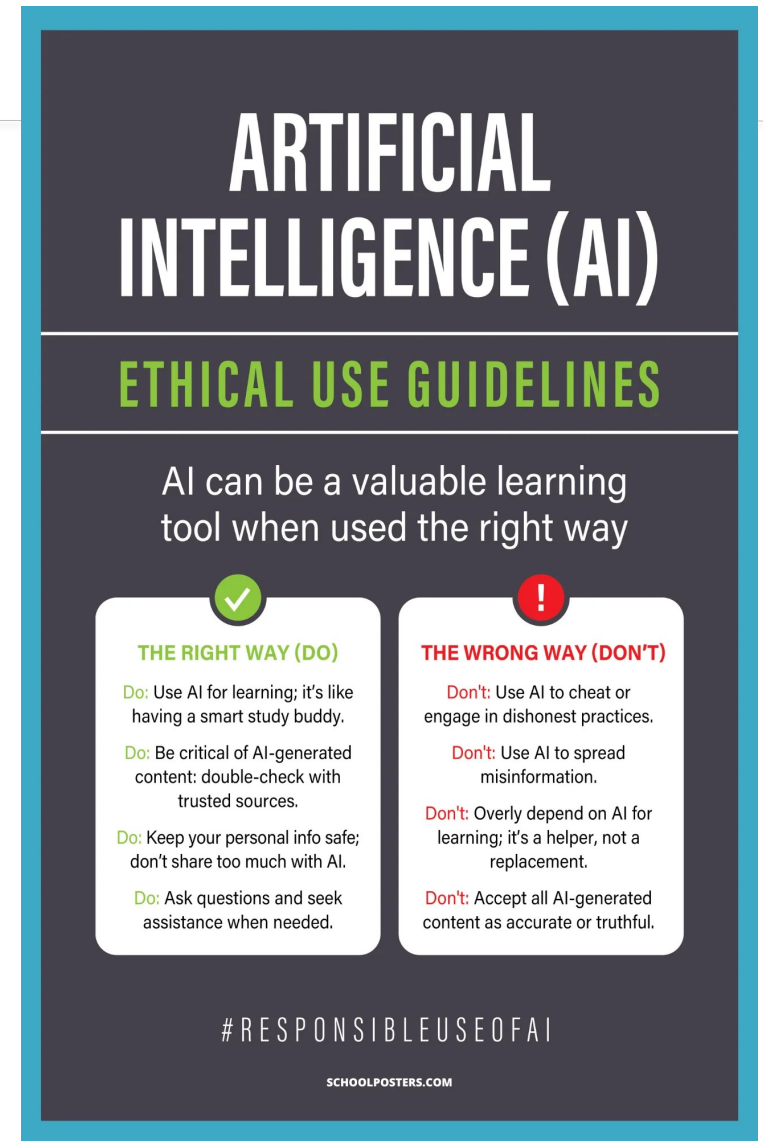
The danger of AI isn't that machines will become too smart—it's that humans might stop thinking

Ethical Use of AI

Good questions to ask before using AI:

- Does this help people or replace them?
- Does this respect agency?
- Does this reduce harm or hide it?
- Am I still responsible for the outcome?

“Tools don’t choose good or evil—people do.”



ARTIFICIAL INTELLIGENCE (AI)

ETHICAL USE GUIDELINES

AI can be a valuable learning tool when used the right way

 THE RIGHT WAY (DO)	 THE WRONG WAY (DON'T)
Do: Use AI for learning; it's like having a smart study buddy.	Don't: Use AI to cheat or engage in dishonest practices.
Do: Be critical of AI-generated content: double-check with trusted sources.	Don't: Use AI to spread misinformation.
Do: Keep your personal info safe; don't share too much with AI.	Don't: Overly depend on AI for learning; it's a helper, not a replacement.
Do: Ask questions and seek assistance when needed.	Don't: Accept all AI-generated content as accurate or truthful.

#RESPONSIBLEUSEOF AI

SCHOOLPOSTERS.COM

Biology/Biomedicine is a Data Science

-W. Evan Johnson

Addendum: don't avoid the math!



Agentic framework for complex epidemiological studies

Complex study data

A single concept may live across different forms, visits, variables, and participant records.

A generalized framework is needed

Each study has its own definitions, timing, and data structure, so the workflow must adapt without being rebuild from scratch.

RePORT India as the first application

We are applying the framework to RePORT India as a realistic test cases for complex longitudinal TB data analysis.

Find the right variables



Design the analysis



Implement & debug



Verify the result

Agents for automated workflow – with investigator review

Generalizable

Reviewable

Large-scale analysis

Accessibility

The study-aware research agent (Application + Database)

Local application

Platform shared across database

- LLM (the brain)
- Tools with diverse capabilities
 - reasoning and utilities
 - uploaded-file interpretation
 - database discovery and extraction
 - statistical execution and result review
- Secure environment (Isolate sandbox)

Study package

Study-specific Design

- Detailed variable definitions
- Verified publication summaries
- Study-design metadata

Available Database

- RePORT India (Deidentified)
- NHANES 2017-2018

Human Review at Key Decision Points

The agent can automate routine execution, while investigators retain control over scientific choices.

Research Question

Agent proposes

Cohort, eligibility,
exposure and outcome

Investigator checks

*"Is this population and
question I intended?"*

Variable Selection

Agent proposes

Variables, forms, visits,
and derived definitions

Investigator checks

*"Are these the correct
variables?"*

Analysis Plan

Agent proposes

Model, covariates,
comparisons, and
missing-data strategy

Investigator checks

*"Is this scientifically
appropriate?"*

Results

Agent proposes

Results, diagnostics,
tables, and figures

Investigator checks

*"Do the results make
sense and support the
conclusion?"*

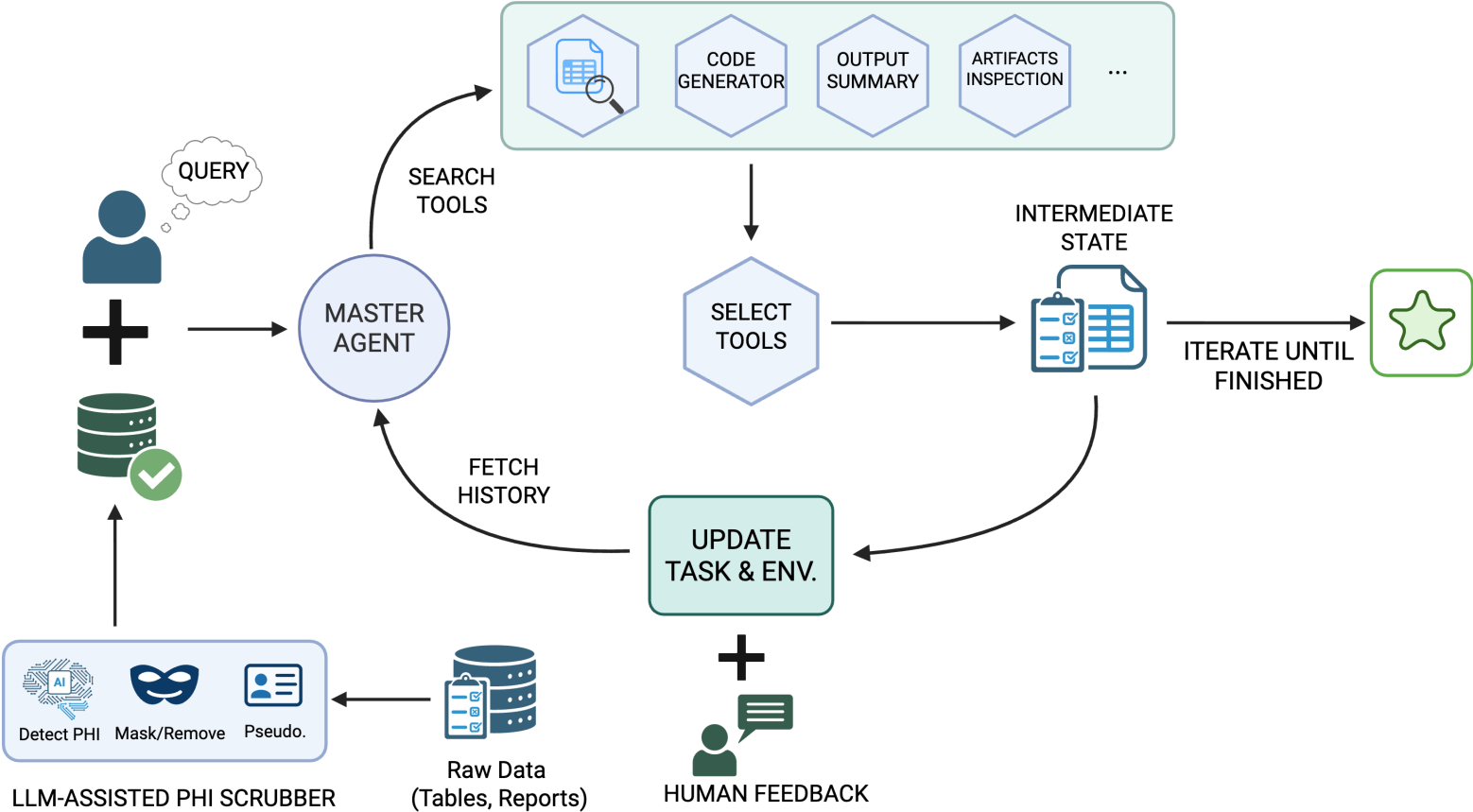
Approve → continue

Revise → agent updates analysis with user feedback

The Framework



The LLM is the single reasoning owner, tools provide observations, and the same loop continues after review or clarification.





ChromeFileEditViewHistoryBookmarksProfilesTabWindowHelp

RePORT Agent

Ask Gemini

127.0.0.1:8000

New Chrome available

<<

Saved conversations

New conversation

AI Agent for RePORT

Ask questions about your data or query from existing database

e.g. Create a baseline index-case dataset with age, sex, marital status, smoking, alcohol use, and diabetes medication use. Give me a simple summary table.

New conversation

gpt-5.6-terra (Medium)

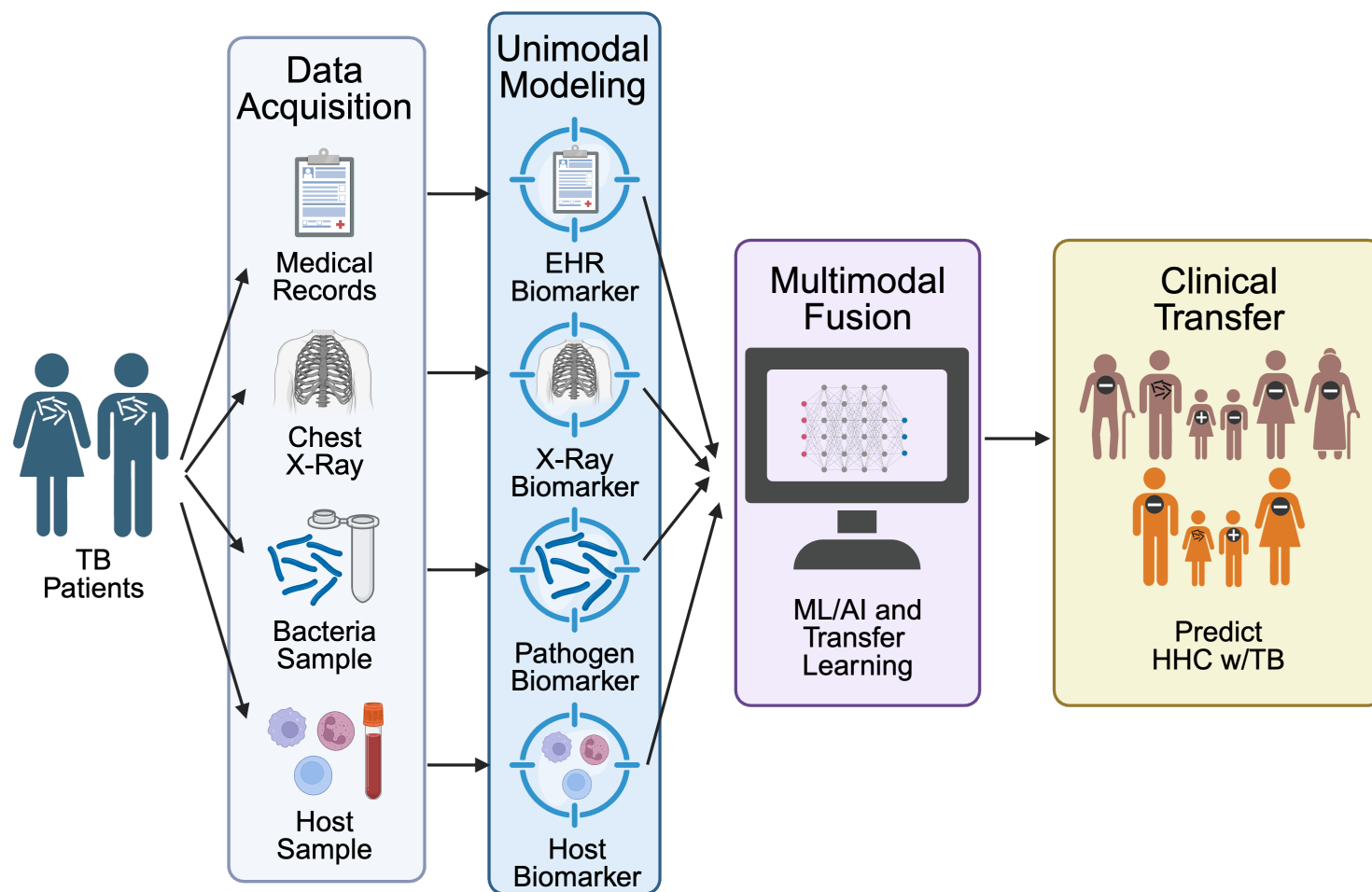
Send

Example Analysis from RePORT India

Study	Dataset	Key variables	Analysis
Zhou et al., 2020; Loss to Follow-up (LTFU) among men; Puducherry, Tamil Nadu	Baseline index case forms; Treatment adherence records; Final treatment outcomes	Sex, marital status, alcohol use, diabetes, BMI, smoking, LTFU	Male subset, multivariable analysis
Van Ness et al., 2017; Delayed care seeking; Puducherry, Cuddalore, Villupuram	Enrollment questionnaire; symptom history; TB knowledge and first-care forms	Cough duration, income, alcohol use, TB knowledge, first-care site	Delay ≥ 4 weeks; multivariable analysis
Ramakrishnan et al., 2021; Mortality during treatment; Three South India districts	Baseline index case forms; hemoglobin/lab data; Final treatment outcomes	Age, sex, income, alcohol use, tobacco, severe anemia, death during treatment	Log-binomial regression; adjusted RR



RUTGERS Predicting TB Transmission





Thank you!